EXPERIMENT NO: 2

Aim: To create and analyze different types of visualizations such as Line Chart, Bar Chart, and Tree Map using the BPB Publication dataset in Microsoft Power BI.

Prerequisite:
Students should have knowledge of:

- Basic computer operations.
- Microsoft Power BI Desktop.
- Importing datasets into Power BI.
- Understanding rows, columns, and data types.

Pedagogy Used:

- Demonstration Method: The instructor demonstrates how to install, launch, and navigate Power BI Desktop, and how to import data from different file formats.
- Hands-on Learning: Students perform each step themselves, including importing datasets, exploring the interface, and opening Power Query Editor.
- Experiential Learning: Students learn by interacting with real-world datasets and observing how Power BI handles different data sources.

Problem: Creating Basic Charts (Line, Bar, and Tree Map) Using the BPB Publication Dataset

Solution:

Clustered Bar Chart

Used to compare sales, profit, or units sold across different categories.

Example

- Sales by Category
- Units Sold by Region

Line Chart

Used to display trends over time.

Example

- Sales over Date
- Profit Trend by Date

Tree Map

Used to visualize hierarchical data and compare the contribution of different categories.

Example

- Sales Contribution by Category
- Profit Contribution by Region

Step 1: Open Power BI Desktop

1. Launch Microsoft Power BI Desktop.

Step 2: Import Dataset

1. Click **Home** → **Get Data** → **Excel**.
2. Select **BPB_Publication_Dataset_50_Records.xlsx**.
3. Click **Load.**

Step 3: Verify Dataset

Ensure the following fields are available:

- Date
- Book_ID
- Book_Title
- Category
- Author
- Region
- Units_Sold
- Unit_Price
- Sales
- Profit

Part A: Create a Clustered Bar Chart

Objective

Compare **Sales across different Book Categories.**

Steps

1. Select **Clustered Bar Chart** from the Visualizations pane.
2. Drag **Category** to the **Y-axis**.
3. Drag **Sales** to the **X-axis**.
4. Sort the chart in descending order.
5. Turn on **Data Labels**.
6. Rename the chart as:

Total Sales by Book Category

Part B: Create a Line Chart

Objective

Analyze the **Sales Trend over Time**.

Steps

1. Select **Line Chart**.
2. Drag **Date** to the **X-axis**.
3. Drag **Sales** to **Values**.
4. Sort the dates in ascending order.
5. Enable Data Labels.
6. Rename the visualization as:

Daily Sales Trend

Part C: Create a Tree Map

Objective

Visualize the **Sales Contribution of Different Regions**.

Steps

1. Select **Tree Map**.
2. Drag **Region** to the **Group** field.
3. Drag **Sales** to the **Values** field.
4. Apply different colors for each region.
5. Rename the visualization as:

Region-wise Sales Distribution

Formatting the Visualizations

Apply the following formatting:

- Add meaningful chart titles.
- Enable Data Labels.
- Change data colors.
- Format the X and Y axes.
- Increase title font size.
- Add borders and shadows.

Analyze the following:

1. Which **book category** generated the highest sales?
2. Which **region** contributed the maximum sales?
3. How do sales vary over different dates?
4. Which category appears largest in the Tree Map?
5. Are there any noticeable sales trends over time?

Outcomes

After completing this experiment, students will be able to:

1. Import an Excel dataset into Microsoft Power BI.
2. Create **Clustered Bar Chart**, **Line Chart**, and **Tree Map** visualizations.
3. Analyze book sales based on **Category**, **Region**, and **Date**.
4. Compare sales performance across different book categories.
5. Identify sales trends over time using Line Charts.
6. Interpret regional sales contribution using Tree Maps.
7. Apply formatting features to improve the appearance and readability of visualizations.
8. Generate meaningful business insights to support data-driven decision-making.

Use Cases

1. **Publishing Industry**

   Analyze the sales performance of books across different categories and regions to identify best-selling genres.

2. **Sales Performance Monitoring**

   Track daily or monthly sales trends to evaluate business growth and seasonal demand.

Viva Questions

Basic Questions

1. What is Microsoft Power BI?
2. What are the advantages of data visualization?
3. What is the purpose of a Clustered Bar Chart?
4. When should a Line Chart be used?
5. What type of data is represented using a Tree Map?
6. What is the difference between Sales and Profit?
7. Why is the Date field important in a Line Chart?
8. What is the role of Categories in a Bar Chart?

Power BI Questions

9. How do you import an Excel dataset into Power BI?
10. What is the purpose of the Visualizations pane?
11. What are Data Labels?
12. How can you change chart colors in Power BI?
13. Why should data types be verified before creating visualizations?
14. What happens if a numeric field is stored as text?

Analytical Questions

15. Which visualization is most suitable for comparing sales across book categories? Why?
16. Why is a Tree Map preferred over a Pie Chart when there are many categories?
17. How can Line Charts help management identify business trends?
18. Which visualization would you recommend to compare sales across different regions and why?
19. What business decisions can be made using the sales dashboard created in this experiment?
20. How can Power BI dashboards improve decision-making in the publishing industry?